



CYBER.SOHO — Educational Hub

Business Information Systems — Management of Information Systems (IS) Assessment

Module 03 · Week 03 · TASK 03 — Key System Applications for the Digital Age

From the Industry to the Classroom — Building the IT's Future, Today and Together!

 Task facts	
Due	Friday, August 28, 2026 · 23:59 (midnight), Central European Time — the Blackboard server clock is authoritative
Where	The DropBox inside the Assignments section of the course's Blackboard — how-to: Blackboard student guide — submitting assignments
Weight	Task 03 = 10% of the course. Tasks 01 + 02 + 03 together = 3 × 10% = 30% " Continuous Course Assessments ", and they form your final Written Project
Workload	Planned for at least 4 hours of real work (≈ 4 h 55 min in the time budget below)
Format	One ZIP file: BIS_Task03_LASTNAME_Firstname.zip (contents listed in "Package & submit")
Topics covered	Achieving Operational Excellence and Customer Intimacy: Enterprise Applications · E-Commerce: Digital Markets, Digital Goods · Managing Knowledge · Enhancing Decision Making



The big picture — one project, four weeks (read this first)

This course does not collect three unrelated homeworks. The three tasks are **three chapters of ONE ongoing capstone**, the **Written Project: "Digital Transformation Blueprint."** You play a junior IS (Information Systems) consultant. Your client is **one fictional small company that YOU invent** (5–50 employees, selling at least one thing that could be sold physically *or* digitally — a bakery, a game studio, a bike shop, a tutoring service...). You keep **the same company** through the whole course, and every task adds one chapter to its blueprint. The workflow follows the official Course Topics plan:

Week	Course topic (per syllabus)	Your capstone output
1	Information systems in the digital age	Task 01 → Chapter 1: the company, its strategy, why IS matters, ethics
2	Information technology infrastructure	Task 02 → Chapter 2: its technology base — hardware, data, networks, security
3	Key System Applications for the Digital Age	Task 03 (this one) → Chapter 3: its application layer — enterprise apps, e-commerce, knowledge, decisions
4	Building and Managing Systems	Chapter 4 + assembly: the build/change plan; you merge Chapters 1–4 into the final Written Project
—	Final Assessment	Written Project (30%) + Comprehensive Exam — as per the Course Evaluation Requirements

Joining late, or Tasks 01/02 used a different company? Allowed. Adopt ONE company now and add a half-page "catch-up profile" as **Appendix A0** (who they are, what they sell, how many people). Record this under *Deviations & Choices* (see the Open-tools rule below) — documented changes are never punished.

Capstone habits that will save you in Week 4: keep one project folder; never delete old versions (add `_v2`, `_v3`); every screenshot goes into an `evidence/` sub-folder; every number you invent gets one sentence saying it is invented.

Before you start — files and free tools

Task 03 **reuses the four working lab files from the Week 03 lecture package** as your starting base (this is deliberate: in industry you almost always extend running code, you rarely start from zero):

```
Week03_L1_ERP_Order_Flow.py · Week03_L2_Mini_eStore.html ·
Week03_L3_Knowledge_Base.py · Week03_L4_WhatIf_Dashboard.html · (optional)
Week03_L4_Cafe_Decision_Data.xlsx
```

Get them from the Week 03 lecture folder on Blackboard, keep them in one folder, and work on **copies**. Free tools that cover everything: [Python 3](#) (runs the two `.py` labs), any modern browser (runs the two `.html` labs by double-click), a free editor such as [Visual Studio Code](#), and [LibreOffice](#) if you do not own Excel. Everything runs **offline**; only the links in this document need internet.



How Task 03 is graded — ponderation 2 – 1 – 3

The task has three parts weighted 2 : 1 : 3, which means **Part A = 20 points, Part B = 10 points, Part C = 30 points → 60 points total**, then scaled to the task's 10% course weight.

Part	What it is	Points	Time plan	Graded on
A — Extend the labs	4 guided code extensions (A1–A4, 5 pts each)	20	≈ 110 min	it runs (3) + evidence screenshot (1) + your comments in the code (1), per extension
B — Concepts, your words	5 short written answers about YOUR company (2 pts each)	10	≈ 40 min	correct concept use + genuinely your own words
C — Chapter 3	the Written-Project chapter, sections C1–C6	30	≈ 130 min	relevance to your company, correct concepts, evidence, documented choices
Packaging & upload	naming, checklist, DropBox upload	—	≈ 15 min	required for any grade
Total		60	≈ 4 h 55 min	



PART A — Extend the four working labs (*20 points · ≈ 110 minutes*)

General rules for all four: work on a **copy**; add a **comment line before every line you add** (same style as the base files); if a lab refuses to run, read the error message bottom-up — it names the line. Each extension ends with **Evidence**: what to screenshot for your `evidence/` folder.

A1 — ERP (Enterprise Resource Planning): second product + safe restocking (*5 pts, ≈ 30 min*)

Goal: prove you understand "one shared database" by growing it.

1. Open your copy of `Week03_L1_ERP_Order_Flow.py` and find the shared dictionary `ERP_DATABASE`.
2. Add a **second product** (your choice — e.g. a drink): give it stock, an official price, and make sure the silo copies in Part B of the file get the same product with a *deliberately old* price (that is the silo bug you must preserve).

3. Write a new function `restock(product_name, quantity)` that **adds** quantity to that product's stock in the ERP database — but **refuses politely** (prints a clear message, does not crash) when the product does not exist or the quantity is zero/negative.
4. In the demo section, restock your new product with a valid amount, then try `restock("Pizza", 10)` — the refusal message must appear.
5. Run it: `python3` your file. Both reports must still print, now with two products. **Check yourself:** ERP totals stay consistent for both products; the unknown-product restock is refused. **Evidence:** the modified `.py` + one screenshot showing the successful restock and the refusal message.

A2 — E-commerce: a third card with subscription revenue (5 pts, ≈ 25 min)

Goal: turn the lecture's revenue-model table into working numbers.

1. Open your copy of `Week03_L2_Mini_eStore.html` in the editor; find the two product cards and the constants at the top of the script.
2. Add a constant `SUB_PRICE = 3.00` — a **subscription** version of the study guide at \$3.00 per month.
3. Duplicate one card block into a **third card** titled "Subscription (digital)". For the same slider value (treat "copies sold" as *active subscribers*), it must show **MRR (Monthly Recurring Revenue) = subscribers × \$3.00**, the tiny digital cost, and monthly profit.
4. Label the number clearly as **"per month"** — that is the whole point of the model. **Check yourself:** at 50 on the slider, the card shows \$150.00 monthly revenue; at 500, \$1,500.00. **Evidence:** the modified `.html` + screenshots at 50 and at 500, all three cards visible.

A3 — Knowledge base: usage ranking + a smarter search (5 pts, ≈ 25 min)

Goal: operate the knowledge-management value chain, not just recite it.

1. Open your copy of `Week03_L3_Knowledge_Base.py`.
2. Write `top_articles(n)` returning the `n` most-used articles, best first (sort by `times_used`), and print a small "Top articles" list at the end of the demo.
3. The search has a known weakness: the ticket "wifi keeps dropping" finds nothing useful. Fix it **your way**: better keywords on the Wi-Fi (Wireless Fidelity) article, a new article, or a smarter matching idea of your own — the Open-tools rule applies.
4. Add a demo ticket with exactly that sentence; it must now return the right fix. **Check yourself:** the ranking order matches the usage counters; the once-failing ticket now succeeds. **Evidence:** the modified `.py` + one screenshot showing the successful "wifi keeps dropping" search and the Top-articles list.

A4 — Decision dashboard: a seventh input — fixed nightly cost (5 pts, ≈ 30 min)

Goal: feel how fixed costs move a break-even point.

1. Open your copy of `Week03_L4_WhatIf_Dashboard.html`; study how one existing slider is wired to `update()`.

2. Add a seventh slider "**Fixed cost per evening (lighting, heating)**", range \$0–\$60, default **\$25**.
 3. Wire it in: fixed cost joins **total cost**, and the **break-even** must now cover it too — per hour that means $(\text{staff} \times \text{wage} + \text{fixed} \div \text{hours}) \div (\text{spend} - \text{ingredient cost})$.
 4. Play: with all defaults, find the customers-per-hour value where GO flips to NO-GO, and write that number as a comment at the top of your file. **Check yourself:** with fixed cost \$0 the dashboard behaves exactly like the original (break-even ≈ 7.8). **Evidence:** the modified .html + one screenshot of a NO-GO verdict + one sentence in your Part B/C document stating the new break-even at defaults.
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PART B — Five concepts, your words, YOUR company (10 points • ≈ 40 minutes)

Answer in **2–4 sentences each**, in your **own words** (copy-pasting lecture sentences scores zero for that item), always about **your capstone company**. Put the answers in your Chapter 3 document under the heading "Part B".

1. Name one realistic **information-silo** risk in your company and say which single design idea from Lesson 1 removes it.
 2. Choose the **e-commerce type** (B2C (Business-to-Consumer) / B2B (Business-to-Business) / C2C (Consumer-to-Consumer)) and **one revenue model** for your company, and justify both in plain words.
 3. Name one piece of **tacit knowledge** your company would lose if one person left, and describe capturing it using the four value-chain steps *by name*.
 4. Pick one real decision your company faces, classify it (**structured** / **semi-structured** / **unstructured**), and name the system type that supports it (MIS (Management Information System) report, DSS (Decision-Support System), or ESS (Executive Support System)).
 5. **GIGO (Garbage In, Garbage Out):** name one way garbage data could enter your company's systems and one habit that prevents it.
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PART C — Chapter 3 of your Written Project (30 points **• ≈ 130 minutes)**

Deliver **one document** (Chapter3_LASTNAME.pdf or .docx or .md, about 3–4 pages) with exactly these sections:

C1 • Company snapshot (*refresher, $\leq$ half a page*) — who the client is, what it sells, headcount. Copy-adapt from Chapters 1–2 or from Appendix A0.

C2 · Enterprise application plan (6 pts) — which of ERP / CRM (Customer Relationship Management) / SCM (Supply Chain Management) your client should adopt **first**, and why, in beginner-honest terms (cost, habit change, switching costs included). Name **one real vendor or product** you researched yourself and give its **official website link** — for a small client, open-source or small-business options such as [Odoo](#) or [Zoho](#) are fair examples, but *your* researched choice is the graded one (Open-tools rule).

C3 · E-commerce & digital goods (6 pts) — one product your client sells online; its e-commerce type and revenue model (may reuse Part B answers, expanded); a **small table** with *your invented numbers* comparing marginal cost and profit for a physical vs digital version, in the spirit of lab A2.

C4 · Knowledge plan (6 pts) — invent **five knowledge-base articles** your client needs (title + 4–6 keywords + author role for each). Load them into your modified A3 program, run one realistic search, and include the screenshot. Close with two sentences: which value-chain step will be hardest for THIS client, and why.

C5 · Decision model (8 pts) — one **semi-structured** decision your client faces. Adapt your A4 dashboard **or** the workbook `Week03_L4_Cafe_Decision_Data.xlsx` with the client's (invented) numbers. Report: the inputs, the computed break-even, the verdict, plus **one what-if sentence** ("if X rises to Y, the verdict flips because..."). Screenshot required.

C6 · Deviations & Choices + AI & Sources statement (4 pts, mandatory) — two short lists: (a) every place you changed a proposed tool, file, technique or example — **WHAT you changed and WHY**; (b) your sources with links, and a one-paragraph statement of any AI (Artificial Intelligence) assistance following the course rule: "**The model drafts. The analyst decides. The analyst's name goes on the record.**" Empty lists are fine — the section itself is not optional.



Package & submit

Build the ZIP `BIS_Task03_LASTNAME_Firstname.zip` containing:

- `Chapter3_LASTNAME.` (pdf|docx|md) — Part B answers + sections C1–C6
- `labs/` — your four modified files: `Week03_L1_*.py`, `Week03_L2_*.html`, `Week03_L3_*.py`, `Week03_L4_*.html` (+ the `.xlsx` if used in C5)
- `evidence/` — the screenshots named `A1.png` ... `A4.png`, `C4.png`, `C5.png`
- (if applicable) `Appendix_A0_company_profile.` (pdf|md)

Upload to the **DropBox in the Assignments** section of the course Blackboard **before Friday, August 28, 2026, 23:59 CET (Central European Time)**. Verify the upload finished — a named receipt/confirmation on screen — before logging out; late or failed submissions follow the course syllabus policy.

 Plain-words dictionary (for newcomers — no prior IS/IT knowledge assumed)

Term	Plain meaning
IS / IT	Information System(s) / Information Technology — the systems and the technology they run on
ERP	Enterprise Resource Planning — all departments share ONE central database, so all numbers agree
Information silo	each department keeps its own copy of data; the copies drift apart and disagree
CRM	Customer Relationship Management — one complete record of every customer interaction
SCM	Supply Chain Management — sharing real numbers along the supplier→customer chain
Switching costs	how expensive it is to leave a vendor once everything runs on it
E-commerce	buying and selling over the internet, including payment and marketing around it
B2C / B2B / C2C	who sells to whom: Business-to-Consumer / Business-to-Business / Consumer-to-Consumer
Revenue model	HOW the money comes in: sales, subscription, advertising, freemium, transaction fee, affiliate
Digital good	a product delivered as pure data (PDF (Portable Document Format), song, software)
Marginal cost	the cost of delivering ONE MORE unit — near zero for digital goods
MRR	Monthly Recurring Revenue — subscribers × monthly price
Tacit / explicit knowledge	in a person's head / written down and searchable
Knowledge base	the searchable collection of written-down solutions
Value chain (knowledge)	acquire → store → share → apply, then repeat
Structured / semi-structured / unstructured decision	rule decides / partly calculable + judgment / judgment only
MIS / DSS / ESS	routine reports / what-if models / executive KPI (Key Performance Indicator) dashboards
What-if analysis	change an assumption, instantly see the recomputed outcome
Break-even	the point where money in equals money out

Term	Plain meaning
GIGO	Garbage In, Garbage Out — output quality can never beat input quality

The Open-tools rule, creativity & academic honesty

Everything proposed here is OPEN. Every tool, technique, file, example, vendor and even most of the exercised content may be **replaced by any other suitable and available alternative** — a different editor, a different language for a lab port, a different vendor in C2, your own program instead of a provided base. Two conditions only: **(1)** the learning goal of the part is still met, and **(2)** you record **WHAT you changed and WHY** in section **C6 — Deviations & Choices**. Documented substitutions are welcomed, not penalized. You are explicitly encouraged to be **creative**, to **research your tools thoroughly** before trusting them, and even to **build your own** tools where you can.

Academic honesty: this is **individual** work; discussing ideas is fine, sharing files or text is not. Invent all business data — never use real personal data (see Canada's federal privacy law PIPEDA (Personal Information Protection and Electronic Documents Act) at the [Office of the Privacy Commissioner of Canada](#) for why that matters). AI assistance is governed by one rule, declared in C6: **the model drafts, the analyst decides, the analyst's name goes on the record** — you verify, you sign, you answer for it.

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